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| <b>SUBJECT:</b><br><br>APPLICATIONS DEVELOPMENT THEORY<br>(APT470S)<br><br>Full time and part time | <b>EXAM DATE:</b><br><br>7 November 2023 | <b>DURATION:</b><br><br>2.5 Hours | <b>TOTAL MARKS:</b><br><br>80 |
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**Cape Peninsula**  
**University of Technology**  
creating futures

**DEPARTMENT OF INFORMATION TECHNOLOGY**  
**FACULTY OF INFORMATICS AND DESIGN**

**COURSE(S):**      **ADICTA: ADVANCED DIPLOMA IN INFORMATION AND COMMUNICATION**  
**TECHNOLOGY: APPLICATIONS DEVELOPMENT**

**EXAMINER:**                      **MS G KHAN**  
**MODERATOR:**                **MR T MAKHURANE**  
**EXTERNAL MODERATOR:** **DR M NORMAN**

### **GENERAL INSTRUCTIONS**

- You are required to answer all questions.
- All test questions relate to the scenario supplied below. Read it carefully before answering the questions.
- This test **counts 30%** of your final mark for this subject and covers work from terms 3 and 4 as per the topic list in the Subject Guide.

## Scenario for *Software Experts*

**Le Buffet** is a prestigious restaurant and it has been one of the trendiest places to dine in Cape Town for two decades. It is well known for its outstanding meals, pairings with drinks, scenic setting and exceptional service. According to the restaurant owner, much of its excellence can be attributed to its “*top-notch people and technology*”. In keeping with the company’s modern outlook, it has invested in several technologies over the years to support and automate its business operations. Currently, several aspects of its systems are under review as the software consultancy named **Software Experts** is assisting **Le Buffet** to take stock of its applications and infrastructure so that they can be improved further. As the chief developer at **Software Experts**, you need to attend to the situations indicated below.

### QUESTION 1 (20 MARKS)

At **Software Experts**, quality is of the utmost importance in both the products that are produced and the processes used to achieve it. Several quality activities are planned to assist with this.

|     |                                                                                                                                                                                    |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1.1 | The first activity under consideration is a technical review. Explain why you think technical reviews are still a relevant practice in current software engineering projects.      | 5  |
| 1.2 | You believe that Formal Technical Reviews (FTR) are effective only if everyone has prepared in advance. Explain how you would recognize a review participant who has not prepared. | 10 |
| 1.3 | Since some of the participants in a FTR are not IT people explain what the outcomes could be of such a meeting and how they will know that it has been finalized.                  | 5  |

### QUESTION 2 (20 MARKS)

|     |                                                                                                                                                                                                                                                                                                                                                     |    |
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| 2.1 | It has been decided that validation tests should be undertaken next. Should these be performed by the developer of the software or the user? Justify your answer and include one example of a test that that person will carry out.                                                                                                                 | 7  |
| 2.2 | You are discussing the importance of having a testing strategy during quality assurance with the new interns at <b>Software Experts</b> . Provide three key reasons that you would include in your explanation.                                                                                                                                     | 3  |
| 2.3 | One of the restaurant’s web apps is being converted to a mobileApp. The developers have incorporated a broader spectrum of user interaction mechanisms to improve the user experience. Tests are now needed for input arising from gesture, and voice input. However, these input mechanisms provide challenges to testing. Explain why this is so. | 10 |

### QUESTION 3 (20 MARKS)

| 3.1                                          | <p>Decisions made on development technologies are known to influence security. Based on the design decisions shown below that were made by the developers of the BOOKINGS app provide two examples of a vulnerability that could arise as a result of it and one example of a control that could be used to offset that vulnerability.</p> <table><tr><th>Technology choice</th><th>Design decision</th><th>Potential Vulnerability</th><th>Control</th></tr><tr><td>Login/password authentication</td><td>System users are authenticated using a login name/password combination. No other authentication method is supported.</td><td></td><td></td></tr><tr><td>Client/server architecture using web browser</td><td>The system architecture is client–server, with clients accessing data through a standard web browser on a client computer.</td><td></td><td></td></tr><tr><td>Use of editable web forms</td><td>Information is presented to users as an editable web form. They can change information in place and upload the revised information to the server.</td><td></td><td></td></tr></table> | Technology choice       | Design decision | Potential Vulnerability | Control | Login/password authentication | System users are authenticated using a login name/password combination. No other authentication method is supported. |  |  | Client/server architecture using web browser | The system architecture is client–server, with clients accessing data through a standard web browser on a client computer. |  |  | Use of editable web forms | Information is presented to users as an editable web form. They can change information in place and upload the revised information to the server. |  |  | 9 |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------|-------------------------|---------|-------------------------------|----------------------------------------------------------------------------------------------------------------------|--|--|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--|--|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---|
| Technology choice                            | Design decision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Potential Vulnerability | Control         |                         |         |                               |                                                                                                                      |  |  |                                              |                                                                                                                            |  |  |                           |                                                                                                                                                   |  |  |   |
| Login/password authentication                | System users are authenticated using a login name/password combination. No other authentication method is supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |                 |                         |         |                               |                                                                                                                      |  |  |                                              |                                                                                                                            |  |  |                           |                                                                                                                                                   |  |  |   |
| Client/server architecture using web browser | The system architecture is client–server, with clients accessing data through a standard web browser on a client computer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |                 |                         |         |                               |                                                                                                                      |  |  |                                              |                                                                                                                            |  |  |                           |                                                                                                                                                   |  |  |   |
| Use of editable web forms                    | Information is presented to users as an editable web form. They can change information in place and upload the revised information to the server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |                 |                         |         |                               |                                                                                                                      |  |  |                                              |                                                                                                                            |  |  |                           |                                                                                                                                                   |  |  |   |
| 3.2                                          | <p>Web and mobileApp development projects are based on agility and this makes it difficult to maintain the usual sequence for managing change. Explain how a continuous stream of changes to content and functionality would have to be managed for such projects being developed for <b>Le Buffet</b>. You may support your answer with a diagram</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 11                      |                 |                         |         |                               |                                                                                                                      |  |  |                                              |                                                                                                                            |  |  |                           |                                                                                                                                                   |  |  |   |

#### **QUESTION 4** (20 MARKS)

|     |                                                                                                                                                      |    |
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| 4.1 | <p>Explain why <b>Software Experts</b> would opt for “pricing to win” as an approach when estimating the price of software products or projects.</p> | 10 |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|----|

4.2

Examine the task durations for the software project activities scheduled below and draw the associated bar graph for this Work Schedule.

10

| Task | Effort (person-days) | Duration (days) | Dependencies  |
|------|----------------------|-----------------|---------------|
| T1   | 15                   | 10              |               |
| T2   | 8                    | 15              |               |
| T3   | 20                   | 15              | T1 (M1)       |
| T4   | 5                    | 10              |               |
| T5   | 5                    | 10              | T2, T4 (M3)   |
| T6   | 10                   | 5               | T1, T2 (M4)   |
| T7   | 25                   | 20              | T1 (M1)       |
| T8   | 75                   | 25              | T4 (M2)       |
| T9   | 10                   | 15              | T3, T6 (M5)   |
| T10  | 20                   | 15              | T7, T8 (M6)   |
| T11  | 10                   | 10              | T9 (M7)       |
| T12  | 20                   | 10              | T10, T11 (M8) |

\*\*\*\*\* END OF PAPER \*\*\*\*\*